



**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD – 402 103
Summer Semester Examination, May 2018**

Branch: FIRST YEAR B. PHARMACY

Subject with Subject Code: BIOCHEMISTRY [BP203T]

Date: 26 / 05 / 2018

Semester: II

Marks: 75

Time: 3 Hrs.

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q. No.1 Multiple Choice Questions: (1 X 20 = 20) (20 Marks)

- 1) Name the compound with greatest free energy
 - a) ATP
 - b) Cyclic AMP
 - c) Phosphocreatine
 - d) Phosphoenolpyruvate
- 2) In TCA cycle Acetyl CoA condenses with _____ to form citrate.
 - a) Acetyl CoA
 - b) Oxaloacetate
 - c) Acetic Acid
 - d) Citric Acid
- 3) A lipid-soluble benzoquinone with a long isoprenoid side-chain
 - a) Ubiquinone
 - b) Cytochrome b
 - c) Cytochrome c
 - d) Cytochrome a
- 4) The cellular organelles called "suicide bags" are_____.
 - a) Lysosomes
 - b) Ribosomes
 - c) Nucleolus
 - d) Golgi's bodies
- 5) Which of the following is a stop codon_____.
 - a) UAG
 - b) UAA
 - c) UCA
 - d) all of these
- 6) Which of the following is a phospholipid .
 - a) Lecithin
 - b) Cholesterol
 - c) Succinic acid
 - d) Carotenoid
- 7) Which of the following is an essential amino acid?
 - a) Valine
 - b) Glycine
 - c) Alanine
 - d) Serine
- 8) The ETC located in_____.
 - a) Ribosome
 - b) Cytoplasm
 - c) Mitochondria
 - d) Nucleus

9) Million Reaction is specific for the amino acid

- a) Valine
- b) Tyrosine
- c) Phenylalanine
- d) Arginine

10) One of the following is absent in DNA structure

- a) Adenine
- b) Guanine
- c) Cytosine
- d) Uracil

11) Edman's reagent contains

- a) Phenylisothiocyanate
- b) 1-Fluoro-2, 4-dinitrobenzene
- c) Urea
- d) Dansyl chloride

12) The process of transfer of amino group from an amino acid to keto acid called as

- a) Oxidative amination
- b) Nonoxidative amination
- c) Transcarboxylation
- d) Transamination

13) Gluconeogenesis can proceed from all of the following except _____

- a) Pyruvate
- b) Palmitic Acid
- c) Propionyl CoA
- d) Lactate

14) Codons are present on

- a) mRNA
- b) tRNA
- c) rRNA
- d) None of these

15) Which of the following is an essential fatty acid?

- a) Oleic Acid
- b) Carboxylic Acid
- c) Cysteinic acid
- d) Salicylic acid

16) Trypsin is an example for the class of enzyme namely

- a) Oxidoreductases
- b) Transferases
- c) Hydrolases
- d) Ligases

17) The major site of urea synthesis is

- a) brain
- b) liver
- c) kidneys
- d) muscles

18) Degree of unsaturation of fats and oils is denoted by _____.

- a) Iodine Number
- b) Phenol Co efficient
- c) Acid Number
- d) Acetyl number

19) Which of the following is not a ketone body?

- a) acetone
- b) acetoacetate
- c) beta hydroxyl butyrate
- d) succinate

20) Rancidity of fat is prevented by addition of _____.

- a) Vit E
- b) Vit A
- c) oxygen
- d. all of these

Q. No 2. Attempt any SEVEN of the following: (7 X 5 = 35)

(35 Marks)

- A) Explain Pentose Phosphate Pathway.
- B) Write note on decarboxylation of amino acid.
- C) What is Glycogenesis? Explain the steps involved.
- D) Write note on transamination reaction. Give importance of SGPT and SGOT.
- E) Differentiate between DNA and RNA. What are Okazaki pieces?
- F) Explain in short ketogenesis process.
- G) Explain urea cycle in detail.
- H) Give in short about factors affecting enzyme activity.
- I) Define and classify lipids with example.

Q. No 3. Attempt any TWO of the following (2 X 10 = 20)

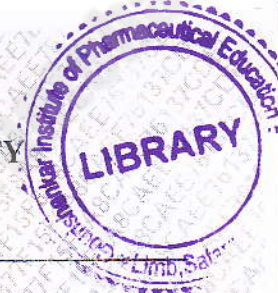
(20 Marks)

- A) Write in detail about β oxidation of fatty acids.
- B) Explain in detail oxidative phosphorylation. Write inhibitors of ETC and uncouplers of oxidative phosphorylation.
- C) What is Glycolysis? Explain the stepwise reactions of glycolysis with generation of ATP.

□□□□□□

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE – RAIGAD – 402 103

Summer End Semester Examination – May 2019



Course: B. Pharm. First Year

Semester: II

Subject: Biochemistry

Subject Code: (BP203T)

Date: 27 /05/2019

Time: 10.30am – 1.30 pm

Marks: 75

- Instructions:** i) All questions are compulsory
 ii) Figures to the right indicate full marks
 iii) Draw the diagrams or flow charts wherever necessary.

Q.No.1 Attempt the following questions (All Questions Compulsory) (20 Marks)

- Oxidation of one molecule of Glucose produce molecule of Pyruvate.
 A) 1 B) 2 C) 4 D) 8
- The amount of energy receives from one molecule of ATP is
 A) 7.3 Kcal B) 76 Kcal C) 760 Kcal D) 100 Kcal
-enzyme catalyses conversion of Fructose 1, 6 diphosphate to Glyceraldehyde -3-Phosphate.
 A) Hexokinase B) Enolase C) Aldolase D) Phosphotrioses Isomerase.
- Isomer formed as result of interchanging position of -H & -OH on carbon 2, 3 & 4 of glucose are known as
 A) Anomer B) Epimer C) Metamer D) D & L isomer.
- Phenomenon of disorganization of native structure of protein is known as
 A) Precipitation B) Denaturation C) coagulation D) oxidation
- Saturated fatty acid is
 A) Not required in diet B) No double bond in structure
 C) Also called as Non essential fatty acid D) All of the above
- Actual amount of energy available to perform chemical reaction is called as.....
 A) Enthalpy change B) free Energy C) Free Energy Change D) Entropy Change.
- The process of synthesis of glucose from non carbohydrate source are called as
 A) Glycolysis B) Gluconeogenesis C) Glycogenesis D) Glycogenolysis.

9. Deficiency of Homogentisate oxidase enzyme in tyrosine metabolism leads todisorder.

- A) Albinism B) Tyrosinemia C) Phenylketonuria D) Alkaptonuria

10. Protein part of conjugated enzyme is called as -

- A) Apoenzyme B) Coenzyme C) Holoenzyme D) exoenzyme

11. Enzyme which are produced in response to substrate are called as -

- A) Repressible enzyme B) Constitutive enzyme C) Induced enzyme

D) Conjugate enzyme

12. Value of Michaelis-Menten Constant (K_m) is calculated with the help of -

- A) Substrate concentration and reaction velocity plot B) Lineweaver -Bunk plot

C) Eadie-Hofstee transform D) All of the above

13. Inhibitor form strong covalent bond with enzyme and permanently inactivate enzyme are called as

- A) Irreversible inhibitor B) Reversible inhibitor C) Non specific inhibitor

D) All of the above

14. In the Biosynthesis of cholesterol, the rate limiting enzyme is

- A) Mevalonate Kinase B) HMG CoA Synthase C) HMG CoA Reductase

D) Cis -prenyl transferase.

15) What is the composition of Nucleotide

- A) Sugar, Phosphate and Amino Acid B) Sugar, Phosphate and Nitrogenous Base

C) Glucose, Phosphate and Amino Acid D) Sugar, Phosphate and Lipid

16. The First product of Urea cycle is -

- A) Ornithine B) Carbamoyl Phosphate C) Citrulline D) Arginosuccinate

17. Which of the following amino acid is precursor of thyroidal hormones, catecholamine and melanin.

- A) Tryptophan B) Serine C) Methionine D) Tyrosine

18. According to Chargaff's rule, in DNA molecule-

- A) The amount of adenine & Thymine is equal to amount of guanine & Cytosine

- B) The amount of adenine & Guanine is equal to amount of Thymine & Cytosine
 C) The amount of adenine & Uralic is equal to amount of guanine & Cytosine
 D) The amount of adenine & Guanine is equal to amount of Uracil & Cytosine

19. In human end product of purine catabolism is

- A) Urea B) Purine oxide C) Uric Acid D) xanthine

20) Hyperglycemia is caused by

- A) Hyperthyroidism B) Hyperpituitarism C) Hyperactivity of pituitary gland
 D) All of the above

Q. No.2. Attempt Any Two of the following

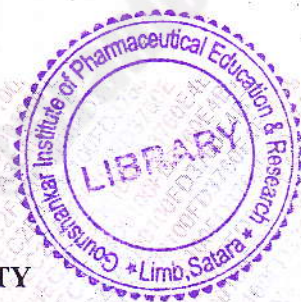
20 Marks

1. Define and classify carbohydrate with suitable example. Explain in detail about TCA cycle along With energetic.
2. Define and Classify enzyme with suitable example. Explain reversible and irreversible enzyme Inhibitor in detail.
3. Explain General Reaction of amino acid with its significance.

Q.No.3. Attempt Any Seven of the following.

35 Marks

1. Write a note on energy rich compound.
2. Write a note on β - oxidation of fatty acid.
3. Define the following terms (**Any Five**)
 a) Epimer b) D & L isomer c) α - amino acid d) Isoelectric point e) saturated fatty acid
 f) Glycogenolysis.
4. What is co-enzyme? Explain structure and function of co-enzyme.
5. Write note on urea cycle & Give its Significance.
6. Write note on structure and Function of DNA.
7. Explain phenomenon of mutarotation with its mechanism.
8. Define enzyme. Explain Factor affecting enzyme activity.
9. Write metabolic disorder of Phenylalanine and tyrosine



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY

LONERE – RAIGAD – 402 103

Winter Semester Examination – December – 2019

Course: B.Pharm.

Subject with Subject Code BIOCHEMISTRY [BP203T]

Date: 07/12/2019

Duration: 3hrs

Semester: II

Marks: 75

Time: 10.30am to 1.30 pm

Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q.No.1 Attempt the following questions (All Questions Compulsory) (20 Marks)

A. Which of the following is an essential amino acid?

- a) Valine
- b) Serine
- c) Alanine
- d) Glycine

B. The main site for oxidative deamination are _____.

- a) Liver & Kidney
- b) Skin & Pancreas
- c) Intestine & stomach
- d) Lung & Skin

C. Name the compound with greatest free energy

- a) ATP
- b) Cyclic AMP
- c) Phosphocreatine
- d) Phosphoenolpyruvate

D. Which nitrogenous base is not found in structure of RNA?

- a) Thymine
- b) Uracil
- c) Guanine
- d) Cytosine

E. The ETC located in _____.

- a) Ribosome
- b) Cytoplasm
- c) Mitochondria
- d) Nucleus

F. The carbohydrate reserved in human body is _____.

- a) Starch
- b) Glucose
- c) Glycogen
- d) Insulin

G. Edman's reagent contains

- a) Phenylisothiocyanate
- b) 1-Fluoro-2, 4-dinitrobenzene
- c) Urea
- d) Dansyl chloride

- H. Million reaction is specific for the amino acid.
- Valine
 - Tyrosine
 - Phenylalanine
 - Arginine
- I. The cellular organelles called "suicide bags" are _____
- Lysosomes
 - Ribosomes
 - Nucleolus
 - Golgi's bodies
- J. The general formula of monosaccharides is _____
- $C_nH_{2n}O_n$
 - $C_{2n}H_{2n}O_n$
 - $C_nH_{2n}O_{2n}$
 - $C_nH_{2n}O_{2n}$
- K. Glucose-6-phosphatase is deficient in _____
- Von Gierke's disease
 - Pompe's disease
 - Cori's disease
 - McArdle's disease
- L. In TCA cycle Acetyl CoA condenses with _____ to form citrate.
- Acetyl CoA
 - Oxaloacetate
 - Acetic Acid
 - Citric Acid
- M. Which of the following is not a ketone body?
- acetone
 - acetoacetate
 - beta hydroxyl butyrate
 - succinate
- N. _____ is *NOT* the factor affecting the enzyme activity.
- Concentration
 - pH
 - Temperature
 - Molarity
- O. Rancidity of fat is prevented by addition of _____.
- Vit E
 - Vit A
 - oxygen
 - all of these
- P. Degree of unsaturation of fats and oils is denoted by _____
- Iodine Number
 - Acid Number
 - Phenol Coefficient
 - Acetyl Number
- Q. Which of the following is a stop codon _____.
- UAG
 - UAA
 - UCA
 - all of these

R. Which of the following is an essential fatty acid?

- a) Oleic Acid b).Carboxylic Acid
- c) Cysteinic acid d).Salicylic acid

S. RNA is converted to protein by _____.

- a) Transcription b) Translation
- c) Lipase d) Kinase

T. Degree of unsaturation of fats and oils is denoted by _____.

- a) Iodine Number b) Phenol Co efficient
- c) Acid Number d) Acetyl number

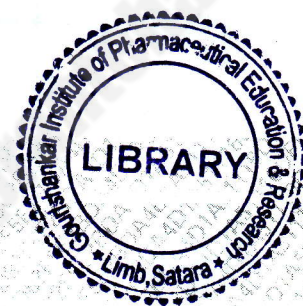
Q.No.2 Attempt any TWO questions of the following: (20 Marks)

- A. What is glycolysis? Explain reaction involved in glycolysis with generation of ATP.
- B. What do you mean by carbohydrate? Give its classification, Discuss in detail its chemical nature and biological role.
- C. Write in detail about β oxidation of fatty acids.

Q.No.3 Attempt any SEVEN questions of the following: (35 Marks)

- ☒ A. Write note on decarboxylation of amino acid.
- B. Define Lipids. Give its classification and add a note on Complex lipids.
- C. Write a note on DNA replication.
- ☒ D. Explain urea cycle in detail.
- ☒ E. Write note on transamination reaction. Give importance of SGPT and SGOT.
- F. Describe factors affecting enzyme activity.
- G. Explain Pentose Phosphate Pathway.
- H. Write chemistry and biological significance of cholesterol.
- ☒ I. Differentiate between DNA and RNA. What are Okazaki pieces?

*****END OF PAPER*****



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: II

Subject Name: Biochemistry

Subject Code: BP203T

Max Marks: 75

Date: 05/09/2022

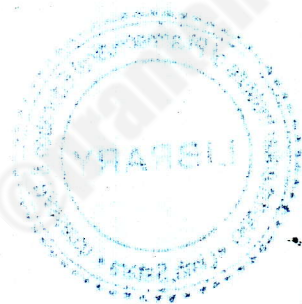
Duration: 3.45 Hr.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = 20 x 1 = 20 (All the questions are compulsory)

- i. _____ enzymes that catalyze the transfer of functional groups.
a) Transferases b) Hydrolases
c) Oxidoreductase d) Lysases
- ii. The two monosaccharides differ from each other in their configuration around a single specific carbon atom, they are referred to as _____ of each other.
a) Tautomer b) Epimer
c) Enantiomer d) Anomer
- iii. Polysaccharides are _____.
a) Oils b) Proteins
c) Fats d) Polymers
- iv. Which one of the following amino acids contains imino group?
a) Tryptophan b) Tyrosine
c) Proline d) Leucine
- v. The cyclopentanoperhydrophenanthrene ring is present in _____.
a) Cholesterol b) Cetyl alcohol
c) Melatonin d) All of the above
- vi. A positive Benedict's test is NOT given by _____.
a) Lactose b) Sucrose



- c) Maltose d) Glucose
- vii. The conversion of alanine to glucose is termed as _____.
a) Glycolysis b) Transformation
c) Glycogenesis d) Gluconeogenesis
- viii. Pentose phosphate pathway generating two important products _____.
a) Pentose sugar and NADPH b) Pentose sugar and NADH
c) Pentose sugars and erythrose sugars d) Pentose sugars and sedoheptulose
- ix. The enzyme _____ deficiency is associated with resistance to malaria.
a) Glucose-6-Phosphate dehydrogenase b) Glucose-6-Phosphate hydrogenase
c) Glucose-5-Phosphate dehydrogenase d) Glucose-5-Phosphate hydrogenase
- x. Obesity is accumulation of _____ in the body.
a) Water b) NaCl
c) Fat d) Protein
- xi. Waxes contain higher alcohols named as _____.
a) Methyl b) Ethyl
c) Heptyl d) Cetyl
- xii. The Enzyme _____ is responsible for black urine disease.
a) Homogentisate oxidase b) Phenyl alanine hydroxylase
c) Tyrosine transaminase d) Carboxylase
- xiii. The transfer of an amino (NH₂) group from an amino acid to a keto acid is known as _____.
a) Transformation b) Transition
c) Transamination d) Deamination
- xiv. In RNA, the complementary base of adenine is _____.
a) Cytosine b) Uracil
c) Guanine d) Thymine
- xv. All are the codons stop the signal in protein synthesis except _____.
a) UAA b) UAG
c) UAC d) UGA
- xvi. The enzyme primase in association with single-stranded binding proteins forms a complex and produces _____.

- a) RNA primer b) RNA polymer
c) DNA primer d) DNA polymer
- xvii. Enzymes lose the catalytic activity at temperature above 70 °C due to _____.
a) Deregulation b) Disturbance
c) Distortion d) Denaturation
- xviii. The reaction given by two or more peptide linkages is _____.
a) Biuret test b) Ninhydrin test
c) Xanthoproteic test d) Pauley's test
- xix. The enzyme _____ is used in the treatment of leukemias.
a) Asparaginase b) Streptokinase
c) Fructokinase d) Hydrogenase
- xx. The functional unit of enzyme is known as _____.
a) Apoenzyme b) Coenzyme
c) Holoenzyme d) Monomeric enzyme

Q. 2. Long Answers = $2 \times 10 = 20$ (Answer 2 out of 3)

- Explain in detail β -oxidation of palmitic acid along with the energetics.
- Define enzyme inhibitors and discuss the different types of enzyme inhibitors.
- Describe the TCA cycle and explain the reaction involved in it with energetics.

Q. 3. Short Answers = $7 \times 5 = 35$ (Answer 7 out of 9)

- Draw the following Structures
a) Glucose b) Tryptophan c) Tyrosine d) Adenine e) Dihydroxy acetone
- Describe the classification of proteins based on their chemical nature with suitable example.
- Define the glycolysis and explain the pathway for glycolysis.
- Give brief account on biological significance and biosynthesis of adrenaline and nor-adrenaline.
- Explain in detail pathway of ornithine cycle.
- Write a note on formation and utilization ketone bodies
- Differentiate between DNA & RNA. Enlist the enzyme involve in DNA replication.
- Define transcription and translation and discuss the inhibitors of protein synthesis.
- Define and Classify enzyme as per IUB system with example.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 11/05/2023

Course : B. Pharmacy

Sem : II

Subject Name : Biochemistry

Subject Code : BP203T

Max Marks : 75 Marks

Duration : 03 Hrs.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 × 2 = 20)

- i) What is OTHLIL? Explain in Short.
- ii) Enlist examples of inhibitors of protein biosynthesis?
- iii) Explain in short about "Ketoacidosis"
- iv) Write full forms of any four
 - a) ATP
 - b) NADH
 - c) NADPH
 - d) FMNH₂
 - e) FAD
- v) Write any four functions of proteins with examples.
- vi) Define & explain in short about Coenzymes.
- vii) Draw Correct double helix structure of DNA
- viii) Mention different metabolic disorders associated with Urea Cycle?
- ix) Write names of any **Eight** enzymes involved in Glycolysis
- x) Explain in short about any two polysaccharides

Q. 2. Long Answers (Answer 2 out of 3)

(2 × 10 = 20)

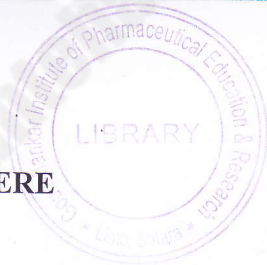
- i) Classify carbohydrates with suitable examples, what are different steps involved in TCA cycle? Explain with its energetics.
- ii) Explain in detail about β -oxidation of fatty acids (Palmitic Acid).
- iii) What are biomolecules? Write definition, classification & biological functions of biomolecules.

Q. 3. Short Answers (Answer 7 out of 9)

(7 × 5 = 35)

- i) Write a note on “Energy Rich Compounds”.
- ii) What are Lipids? Write Classification with examples and biological role of Lipids.
- iii) Write in brief about “Glycogenolysis”.
- iv) Explain about inhibitors of ETC.
- v) Write a short note on “Disorders associated with Tyrosine Metabolism”.
- vi) Explain Stages involved in RNA Transcription.
- vii) Write a short note on Hyperuricemia & Gout disease.
- viii) Explain Enzyme Kinetics with focus on “Effect of concentration of substrate”.
- ix) Write a short note on Diabetes Mellitus.

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2023

Date: 11/09/2023

Course : B. Pharmacy
Subject Name : Biochemistry
Max Marks : 75

Sem : II
Subject Code : BP203T
Duration : 3hr

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Define transamination and deamination.
- ii) Write differences between DNA and RNA.
- iii) Define endergonic and exergonic reactions.
- iv) Give biological significance of cholesterol.
- v) Define nucleosides and nucleotides
- vi) Write the biological functions of proteins.
- vii) Draw structures of Sucrose and Tyrosine.
- viii) Define essential fatty acids with examples.
- ix) What is fatty liver and phenylketonuria?
- x) Define energy rich compounds with examples.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Explain the various reactions involved in the Citric acid cycle with energetics.
- ii) Explain β -oxidation of fatty acid along with the energetics.
- iii) Define Enzyme inhibitors and discuss different types of enzyme inhibitors.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Explain electron transport chain.
- ii) Give the reaction sequence of Glycolysis pathway.
- iii) Define enzymes and classify them according to IUB with examples.
- iv) Define carbohydrates classify them with example and give their biological significance.
- v) What is ketoacidosis? Write the formation of ketone bodies.
- vi) Explain biosynthesis of purine.
- vii) Describe the hormonal regulation of blood glucose level.
- viii) Give the reaction sequence in Urea Cycle.
- ix) Explain replication of DNA.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: II

Subject Name: Biochemistry

Subject Code: BP203T

Max Marks: 75

Date: 18/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10 X 2) = 20

- i) Give a classification of carbohydrates with a suitable example.
- ii) Draw the structure of α D Glucose.
- iii) Which enzyme is involved in the conversion of Glucose -6-phosphate to fructose- 6-phosphate and Glucose -6-phosphate to glucose .
- iv) How is glycolysis regulated?
- v) What are different methods of catabolism of purine nucleotides ?
- vi) Define amino acids. Classify them with examples.
- vii) Enlist steps involved in DNA replication by semi-conservative model.
- viii) Explain the hydrolytic deamination.
- ix) What are holoenzymes and prosthetic groups?
- x) Give classification of Enzymes with an example of each.

Q.2. Long Answers (Answer 2 out of 3) (10 X 2) = 20

- i) Give brief account of TCA cycle with its significance and energetics
- ii) Explain the biosynthesis of 5-HT and melatonin.
- iii) Discuss β -oxidation of saturated fatty acids with energetics.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Give details of bioenergetics of glycolysis.
- ii) Explain in detail bile acid synthesis.
- iii) Write a short note on urea cycle.
- iv) Define polysaccharides and describe the structure of any one polysaccharide.
- v) Explain the different methods of catabolism of amino acids.
- vi) Describe HMP shunt pathway and give its regulation with significance.
- vii) Write a short note on Formation of Melanin.
- viii) Write a short note on ketone bodies.
- ix) Explain the different types of Enzyme inhibitors with suitable examples.

*** END OF THE PAPER ***

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter 2023

Date:08/01/2024

Course : B. Pharmacy

Subject Name: Biochemistry

Max Marks : 75

Sem: II

Subject Code: BP203T

Duration : 3 hrs.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Classify lipids with suitable examples.
- ii) Draw the structures of Sucrose and Starch.
- iii) What are Purine and Pyrimidine bases.
- iv) Draw the double helix structure of DNA.
- v) Define Essential and Non-essential amino acids.
- vi) Define Redox potential and Energy rich compounds.
- vii) What is Gluconeogenesis and HMP shunt.
- viii) Give the difference between DNA and RNA.
- ix) Define Endergonic and Exergonic reactions.
- x) Draw the structures of α -helix and β -Pleated sheet.

Q.2 Long Answers (Answer 2 out of 3) (2x10) = 20

- i) Define and classify Proteins with example. Explain the various reactions involved in Urea cycle and its disorders.
- ii) What are lipids. Write in detail about ketogenesis. Give the disorders of lipid metabolism.
- iii) Define Glycolysis. Explain the pathway of glycolysis with energy production. Write a note on Glycogen Storage disease.

Q.3 Short Answers (Answer 7 out of 9) (7x5) = 35

- i) Give the reaction sequence in Gluconeogenesis.
- ii) Describe the Electron transport chain.
- iii) Explain Transamination in brief.
- iv) Define Enzymes and Co-enzymes. Give the IUB classification with example.
- v) Explain stages involved in RNA Transcription.
- vi) Write a short note on Diabetes Mellitus.
- vii) What is ketoacidosis. Give the formation of Ketone bodies.
- viii) Write a note on Enzyme Inhibition.
- ix) Explain DNA Replication.